

Why is Matrix Multiplication That Way?

FRORDO

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1. Introduction

When I first learned about matrices, I naively expected them to multiply as follows:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 & 2 \cdot 3 \\ 3 \cdot 4 & 4 \cdot 5 \end{pmatrix}$$

Instead, they multiply like

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 + 2 \cdot 4 & 1 \cdot 3 + 2 \cdot 5 \\ 3 \cdot 2 + 4 \cdot 4 & 3 \cdot 3 + 4 \cdot 5 \end{pmatrix}$$

Why is that?

2. Linear Transformations

A transformation takes a vector as input and outputs a vector. More formally, $T : V \rightarrow W$ where V and W are vector spaces. A *linear* transformation L has the following properties:

- $L(a\vec{v}) = aL(\vec{v})$
- $L(\vec{v} + \vec{w}) = L(\vec{v}) + L(\vec{w})$

$$\begin{bmatrix} 1 & 2 & \dots & 10 \\ 2 & 2 & \dots & 10 \\ \vdots & \vdots & \ddots & \vdots \\ 10 & 10 & \dots & 10 \end{bmatrix}$$

$$\begin{pmatrix} | & | & & | \\ b_1 & b_2 & \dots & b_n \\ | & | & & | \end{pmatrix}$$

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$$\left(\begin{array}{c} \boxed{A} \end{array} \right)$$